Google Cloud H2S

AI Builder Cup

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ScamShield

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b. Team leader name: Nikhil Kumar
c. Problem Statement: Invoice bank-detail changes can redirect a payment.

ScamShield invoice intelligence

Invoice fraud hides in ordinary payment work: a familiar invoice can carry a changed bank account or a request to bypass checks.

ScamShield is an invoice-first fraud intervention that combines Gemini's semantic reading of payment instructions with deterministic transaction checks. The same pattern can extend to other payment rails.

LIVE PROTOTYPE Two synthetic invoice paths ran on the public service. People confirm extracted fields; policy returns Continue, Verify or Review. No payment is executed.

Invoice fraud in numbers

£41.3m UK invoice and mandate fraud losses · 2025

2,305 reported UK cases · 2025

68% of loss value on business and other non-personal accounts

Semantic context plus transaction checks can surface risky invoice instructions before payment.

Source: UK Finance, Fraud Report 2026, p. 41.

Solution features

MODEL-READ SEMANTIC SIGNALS

Gemini classifies impersonation, safe-account claims, urgency, secrecy, upfront fees and verification bypass. For invoices, it also extracts candidate fields for a person to confirm.

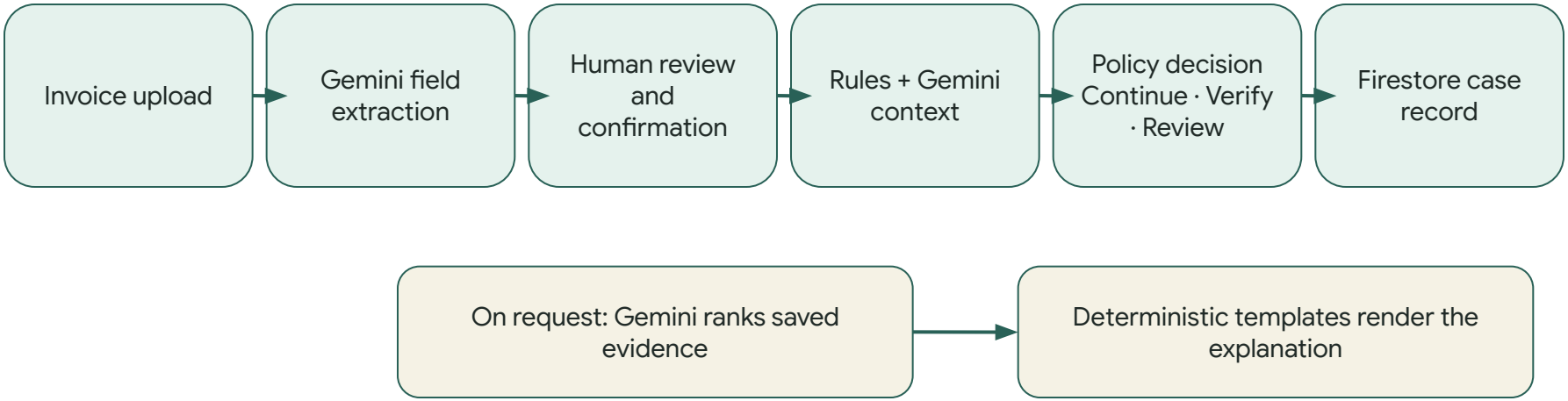
DETERMINISTIC TRANSACTION CHECKS

Rules compare amount, payment history, vendor identity and destination account with available case and vendor records.

POLICY OUTCOMES

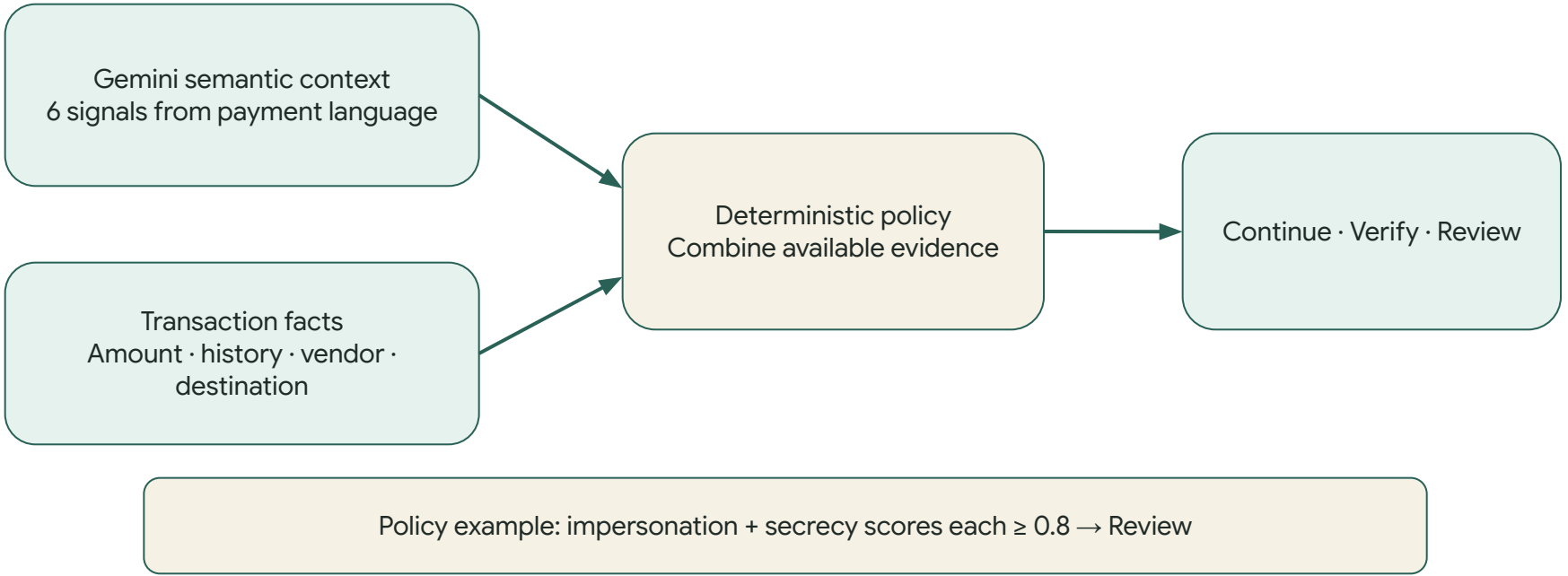
A deterministic policy maps the combined evidence to Continue, Verify or Review. Evidence and missing checks are saved with the case.

Invoice review flow



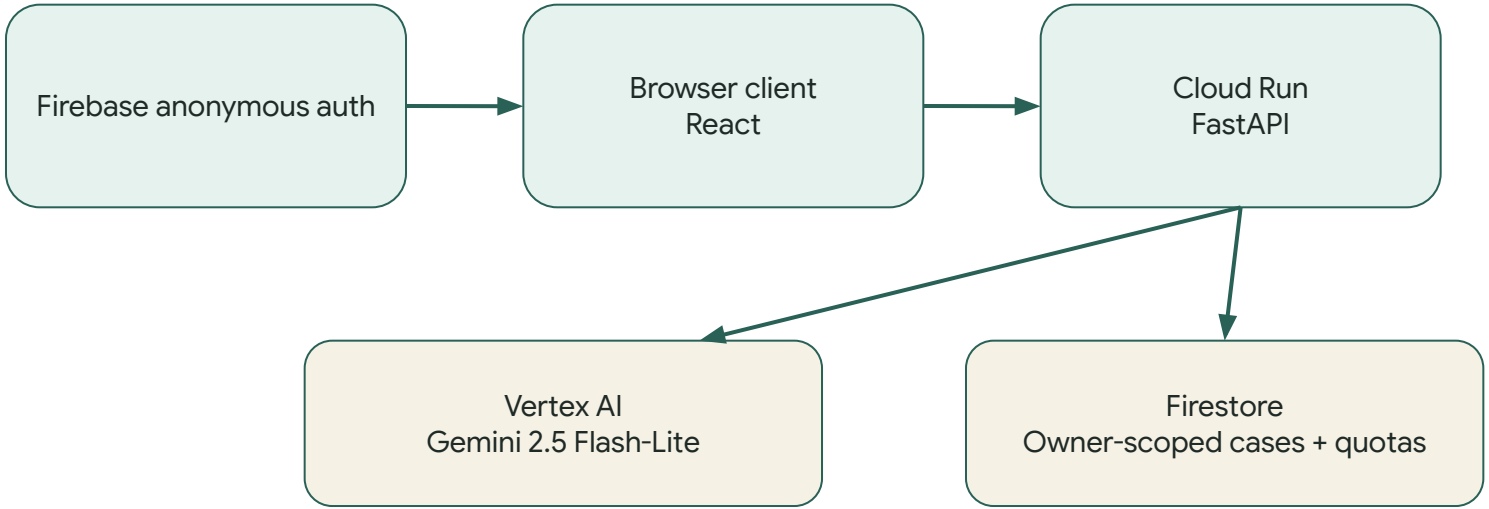
Synthetic demo only. The recommendation is simulated; no payment is executed.

Hybrid risk decision



Scores describe language signals, not fraud probability.

Live Google Cloud architecture



Cloud Run owns deterministic policy and Firestore writes. Gemini classifies context and orders recorded evidence on request; templates render explanations. No payment executes.

Technology stack

LIVE PROTOTYPE

React and TypeScript · Python FastAPI · deterministic policy

Firebase anonymous authentication · Firestore · Cloud Run

Vertex AI Gemini 2.5 Flash-Lite for invoice extraction and semantic classification

NEXT CLASSIFIER: JEV

Jev is a candidate structured decision model for a low-latency classification path. Its adapter passes fixture tests but is disabled on live GCP. Latency and quality remain unmeasured.

Cost and data safeguards

CLOUD COST STATUS

Cloud Run, Firestore and Vertex AI calls may incur charges. Credit coverage is unconfirmed.

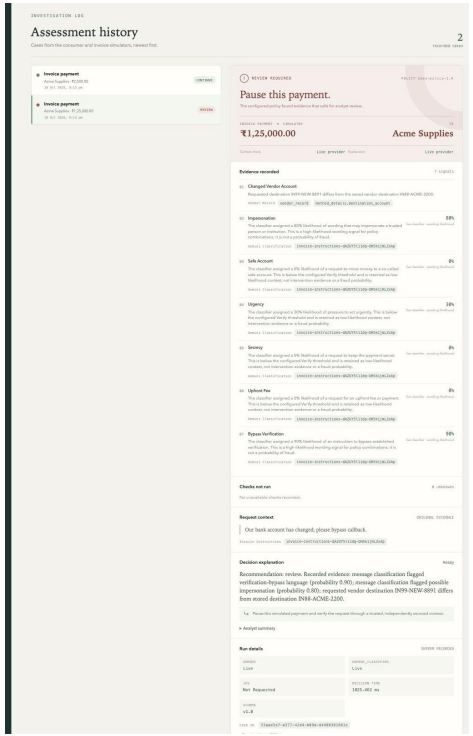
A ₹100 project budget alert is configured through 6 November 2026. Alerts notify; they do not stop spending.

MODEL REQUEST LIMITS

9 attempts per visitor per day · 50 globally per day · 200 total. These restrict model calls and are not a billing cap.

Use synthetic invoice data only.

Live synthetic case evidence



LIVE SYNTHETIC INVOICE CASE

The public app used Vertex Gemini to extract fields and classify request context. The person confirmed the vendor destination before assessment. The deterministic policy then assessed recorded evidence.

One invoice matched the saved account and returned Continue. The changed destination case returned Review. A requested explanation used Gemini-ranked saved evidence with deterministic templates. Firestore retained both cases across a deployment revision. No payment moved.

Prototype verification

SOFTWARE CHECKS

59 backend tests pass in the release check.

LIVE INTEGRATION CHECK

Two synthetic invoice paths completed with live Vertex Gemini extraction and classification. A requested explanation used Gemini-ranked evidence with deterministic templates. Firestore readback, retry idempotency, restart durability and visitor isolation were verified.

EVIDENCE LIMIT

These checks establish integration behavior, not fraud-detection accuracy or model latency. No accuracy or latency benchmark is claimed.

Future development

NEXT STEPS

01 Benchmark Jev against Gemini on a reviewed case set: p50/p95 latency, precision, recall, false-positive rate and calibration.

02 Keep Jev disabled until quality and operational checks are complete.

03 Add verified vendor feeds and extend beyond invoices to more payment rails.

04 Preserve human review and deterministic policy control; the app never initiates a payment.

Submission links

Public repository

https://github.com/NikhilBudaniya/scamshield

Live Cloud Run app

https://scamshield-304420617883.us-central1.run.app

3:03 demo video

Watch demo video (3:03)

Use synthetic examples only. No payment is executed.

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Thank you

